
EE 543: Vision and Learning Based Control

Course Code	EE 543
Course Name	Vision and Learning Based Control
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	3rd/4th year B.Tech. EE, CS, and M.Tech/Ph.D.
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction :** Overview, motivation, and real-world practical applications. [1L]
- **Visual Sensor Model and Calibration:** Camera model, Coordinate Frames and Transforms, Intrinsic camera calibration, and extrinsic camera calibration. [10L]
- **Visual Servoing:** Image Jacobian, Robot Jacobian, Image Based Visual Servoing, Position Based Visual Servoing, Eye-in-hand and Eye-to-hand Configurations, Comparison among different class of visual servoing. [14L]
- **Robot Learning:** Basic concepts of reinforcement learning, reinforcement learning algorithms. Robot learning by demonstration. [14L]
- **Hybrid Method Design:** Comparative analysis for various methods. Explore, understand and identify different ways to design a hybrid scheme to control the given system of interest. Case study and course projects. [3L]

Text Books

1. Corke, Peter I., and Oussama Khatib, **Robotics, vision and control: fundamental algorithms in MATLAB**, Vol. 73. Berlin: Springer, 2011.
2. Vakanski, Aleksandar, and Farrokh Janabi-Sharifi, **Robot learning by visual observation**, John Wiley & Sons, 2017.

References

1. Ijspeert, Auke Jan, et al., **Dynamical movement primitives: learning attractor models for motor behaviors**, Neural computation 25.2 (2013): 328-373.
2. Chaumette, François, and Seth Hutchinson, **Visual servo control I: Basic approaches**, IEEE Robotics & Automation Magazine 13.4 (2006): 82-90.
3. Chaumette, François, and Seth Hutchinson, **Visual servo control III: Advanced approaches [Tutorial]**, IEEE Robotics & Automation Magazine 14.1 (2007): 109-118.